

Exam Answer: Cell Membrane Structure and Function

Student: Variant B (PDF submission)

The plasma membrane, also called the cell membrane, is a flexible boundary that encloses all living cells. It mainly consists of a phospholipid bilayer along with embedded proteins, cholesterol, and carbohydrate molecules.

Its main job is to control what substances enter and exit the cell, which is called selective permeability. Small molecules like oxygen and carbon dioxide diffuse across it easily, but larger molecules need transport proteins to pass through.

The membrane is also important for cell signaling since it lets cells communicate with their surroundings via receptor proteins. It further gives structural support and helps the cell keep its shape.